

class10

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The PDB database

The **Protein Data Bank** (<http://www.rcsd.org/>) is the main repository of biomolecular structure data. Let's see what is in it:

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

X-Ray is 81% and Electron Microscopy is 13%. >Q2: What proportion of structures in the PDB are protein?

```
stats <- read.csv("pdb_stats.csv", row.names = 1)
protein_rows <- grep("^Protein", rownames(stats))
protein_count <- sum(stats[protein_rows, ])
total_count <- sum(stats)
protein_proportion <- protein_count / total_count
protein_proportion
```

```
[1] 0.979118
```

```
stats <- read.csv("pdb_stats.csv", row.names=1)
stats
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	178795	21825	12773	343	226	84
Protein/Oligosaccharide	10363	3564	34	8	11	1
Protein/NA	9106	6335	287	24	7	0
Nucleic acid (only)	3132	221	1566	3	15	3
Other	175	25	33	4	0	0
Oligosaccharide (only)	11	0	6	0	1	0
	Other	Total				
Protein (only)	32	214078				
Protein/Oligosaccharide	0	13981				
Protein/NA	0	15759				
Nucleic acid (only)	1	4941				
Other	0	237				
Oligosaccharide (only)	4	22				

```
n.sums <- colSums(stats)
n <- n.sums/n.sums["Total"]
round(n, digits=2)
```

	X.ray	EM	NMR	Integrative
	0.81	0.13	0.06	0.00
Multiple.methods	Neutron		Other	Total
	0.00	0.00	0.00	1.00

What is the total number of entries in the PDB

```
n.sums["Total"]
```

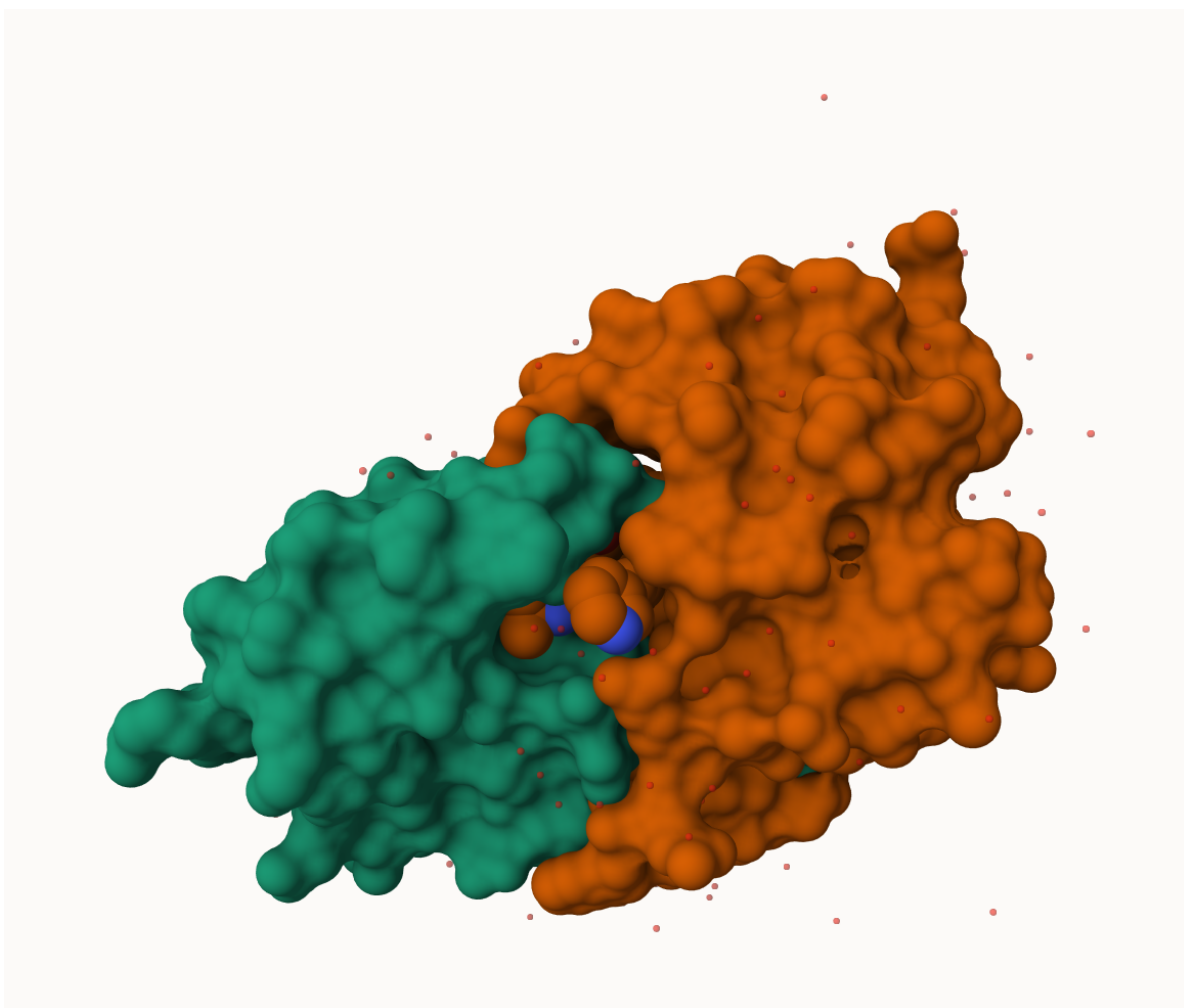
```
Total
249018
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

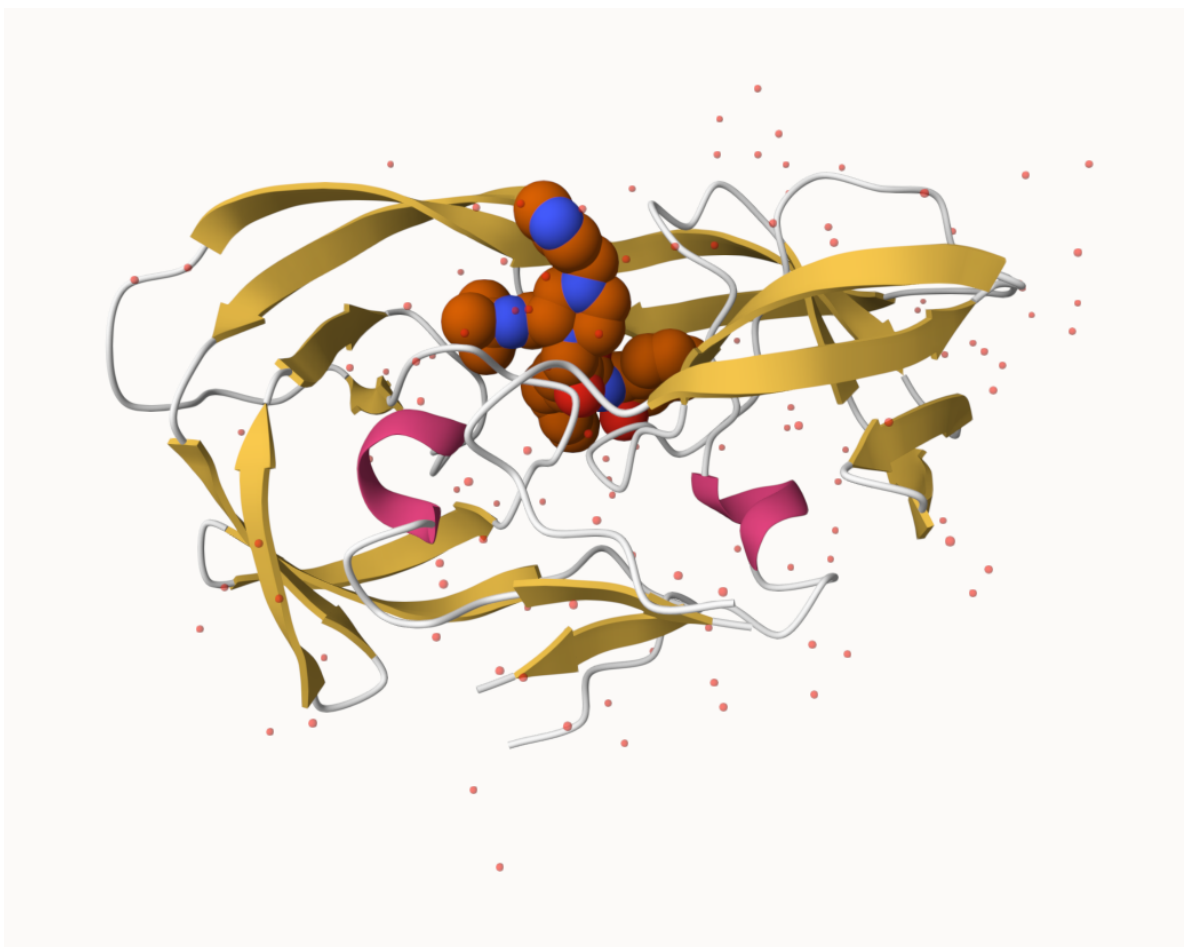
4,940 structures

Using Molstar

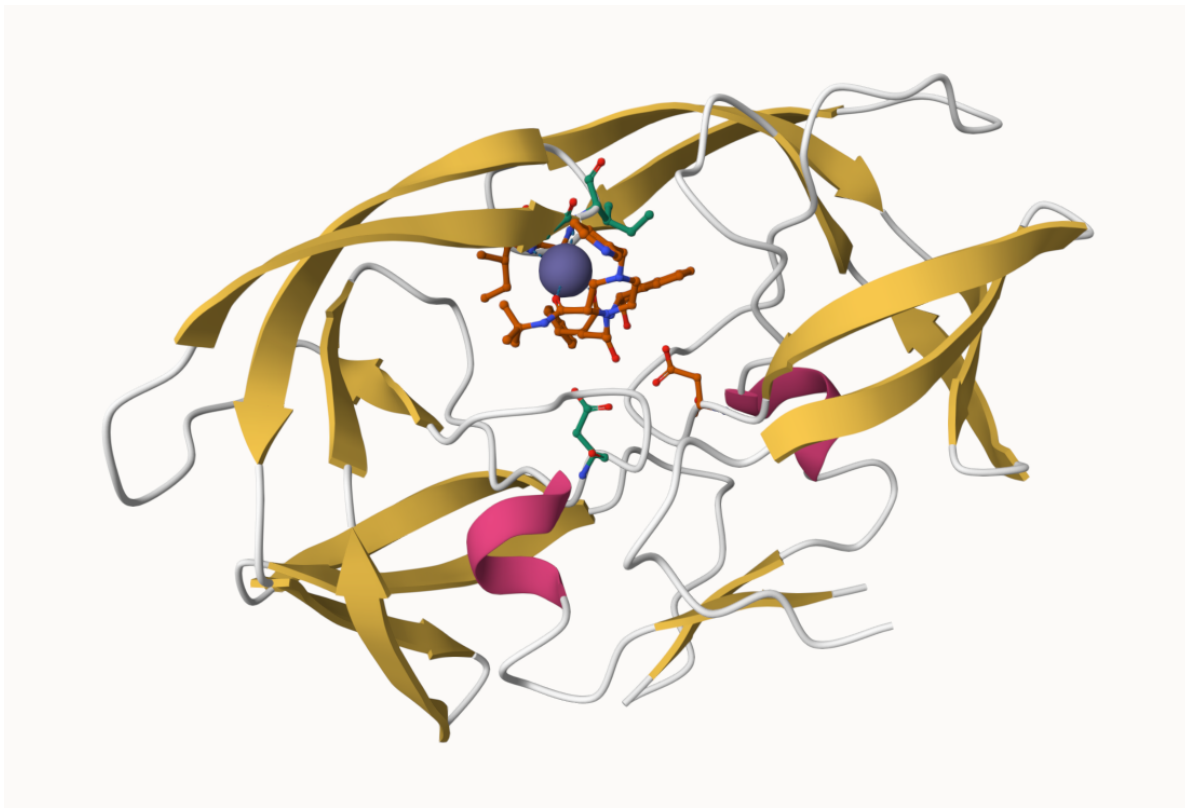
We can use the main [Molstar viewer online](#):



>Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary structure, showing the inhibitor (ligand) in spacefill.



Q. One final image showing catalytic APS 25 and the all important active site water molecule as spacefill.



The Bio3D package for structural bioinformatics

```
library(bio3d)  
hiv <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
hiv
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
 Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
 Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
 QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
 ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
 VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
 calpha, remark, call

`head(hiv$atom)`

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

`pdbseq(hiv)`

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
"P"	"Q"	"I"	"T"	"L"	"W"	"Q"	"R"	"P"	"L"	"V"	"T"	"I"	"K"	"I"	"G"	"G"	"Q"	"L"	"K"
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
"E"	"A"	"L"	"L"	"D"	"T"	"G"	"A"	"D"	"D"	"T"	"V"	"L"	"E"	"E"	"M"	"S"	"L"	"P"	"G"
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
"R"	"W"	"K"	"P"	"K"	"M"	"I"	"G"	"G"	"I"	"G"	"G"	"F"	"I"	"K"	"V"	"R"	"Q"	"Y"	"D"
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80

```

"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99  1
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"

```

Let's try out the new **bio3dview** package that is not yet a CRAN.

```

library(bio3dview)

sele <- atom.select(hiv, resno=25)
#view.pdb(hiv, backgroundColor = "pink",
#         highlight = sele,
#         highlight.style = "spacefill")

```

Prediction of Protein flexibility

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

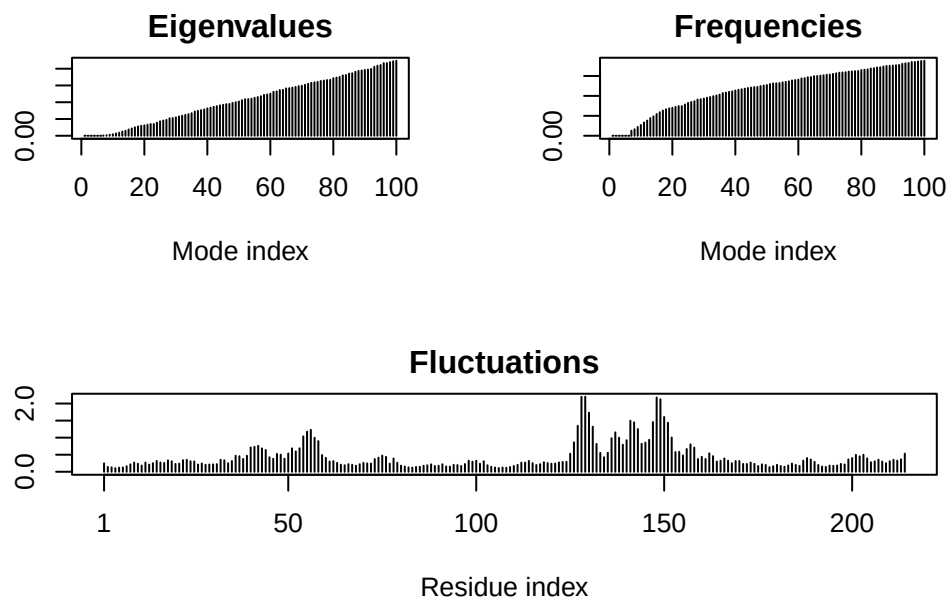
PDB has ALT records, taking A only, rm.alt=TRUE

```
m <- nma(adk)
```

Building Hessian... Done in 0.022 seconds.

Diagonalizing Hessian... Done in 0.081 seconds.

```
plot(m)
```



Write out our results as a wee trajectory movie:

```
mktrj(m, file="results.pdb")
```

```
#view.nma(m)
```